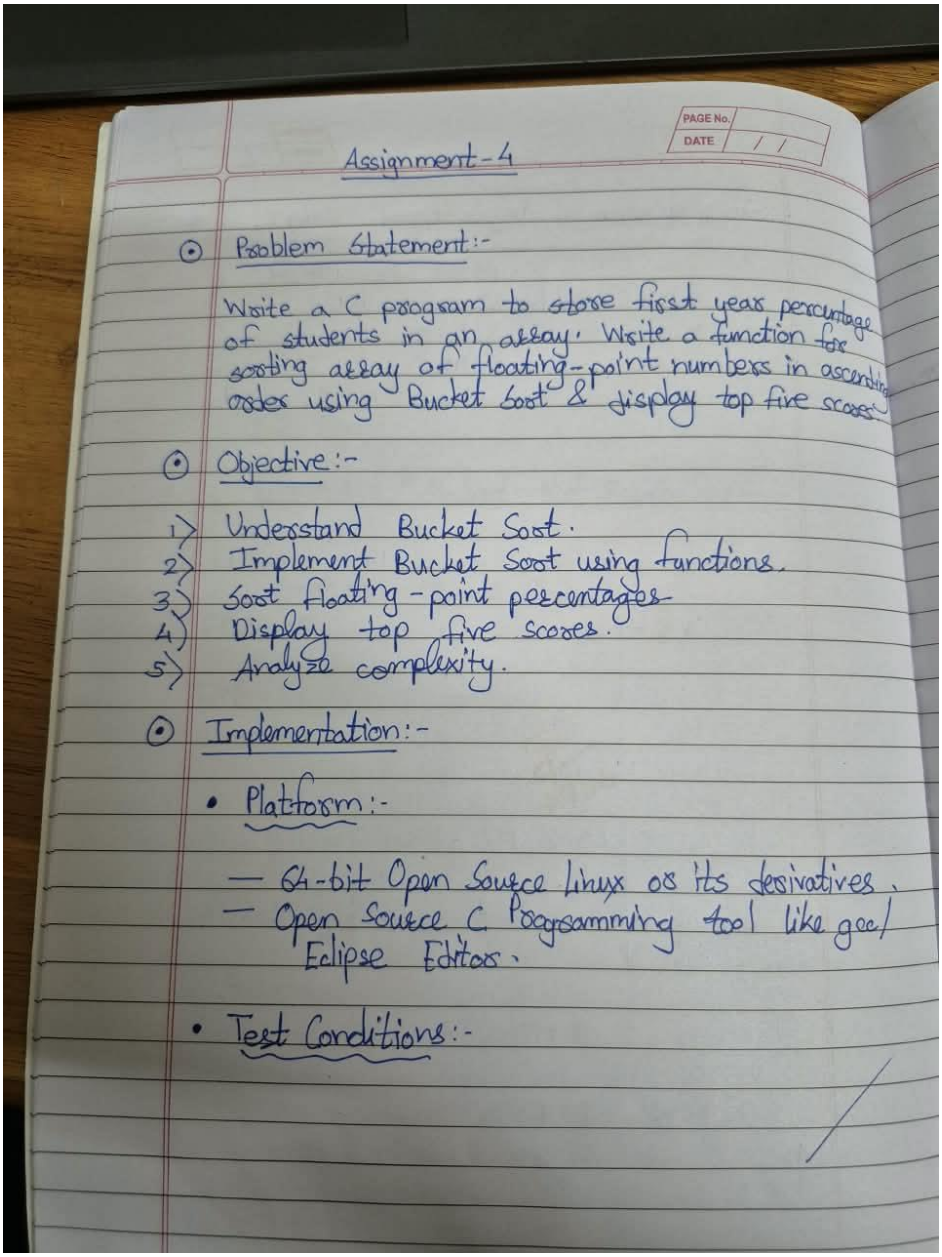
# ASSIGNMENT-4

NAME – Abhinav Ajitkumar Chavan

PRN – 1262251966

BATCH - A1



Assignment-4

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⊙ Problem Statement:-

Write a C program to store first year percentage of students in an array. Write a function for sorting array of floating-point numbers in ascending order using Bucket Sort & display top five scores.

⊙ Objective:-

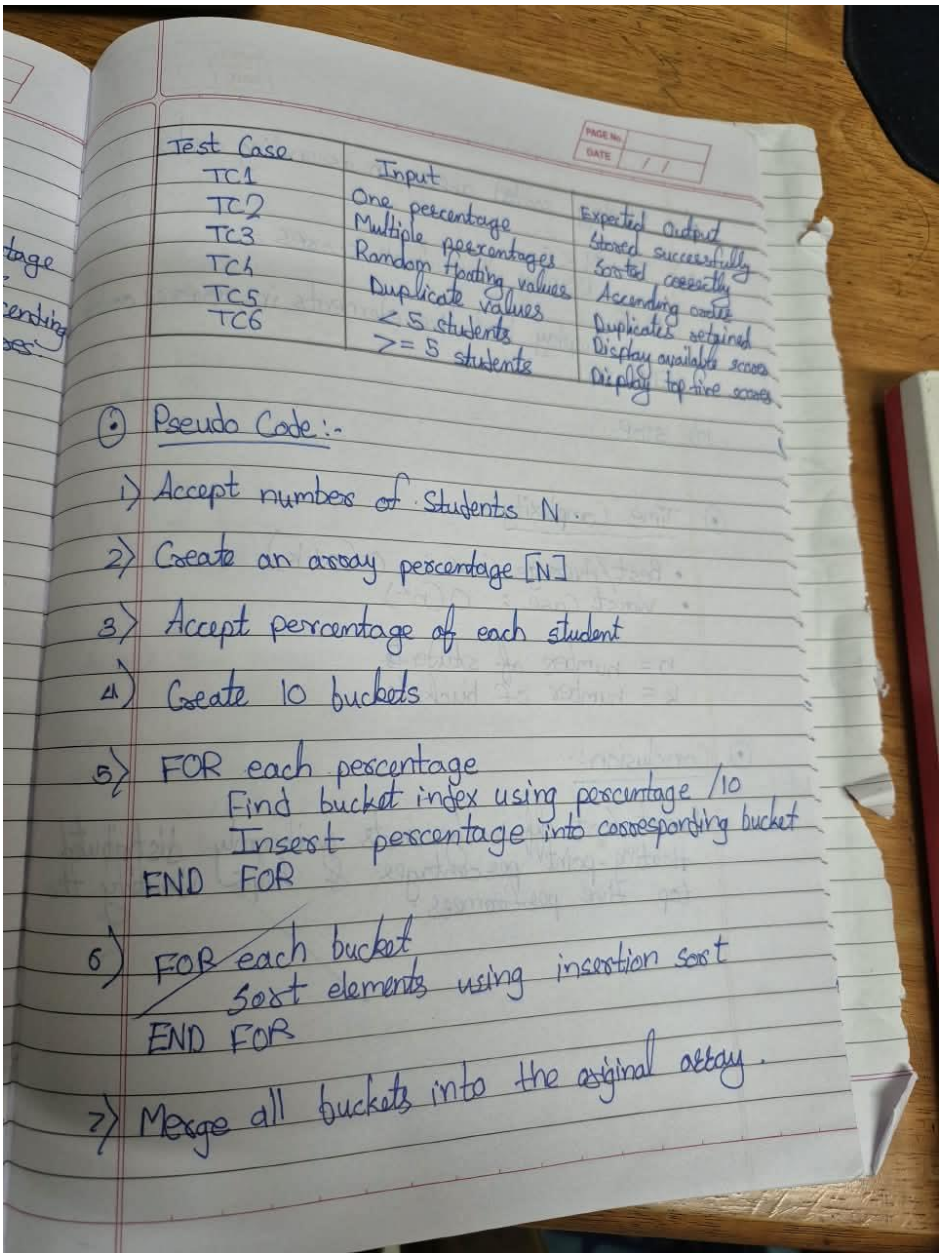
1) Understand Bucket Sort.
2) Implement Bucket Sort using functions.
3) Sort floating-point percentages.
4) Display top five scores.
5) Analyze complexity.

⊙ Implementation:-

• Platform:-

— 64-bit Open Source Linux os its desivatives.
— Open Source C Programming tool like gcc/ Eclipse Editor.

• Test Conditions:-



| Test Case | Input | Expected Output |
|---|---|---|
| TC1 | One percentage | Stored successfully |
| TC2 | Multiple percentages | Sorted correctly |
| TC3 | Random floating values | Ascending order |
| TC4 | Duplicate values | Duplicates retained |
| TC5 | < 5 students | Display available scores |
| TC6 | >= 5 students | Display top five scores |

① Pseudo Code :-

1) Accept numbers of students N.

2) Create an array percentage [N]

3) Accept percentage of each student

4) Create 10 buckets

5) FOR each percentage
   Find bucket index using percentage /10
   Insert percentage into corresponding bucket
   END FOR

6) FOR each bucket
   Sort elements using insertion sort
   END FOR

7) Merge all buckets into the original array.

8) Display sorted array in ascending order

a) IF N < 5
    Display all available scores.
ELSE
    Display last five elements in reverse order.
END IF

10) STOP.

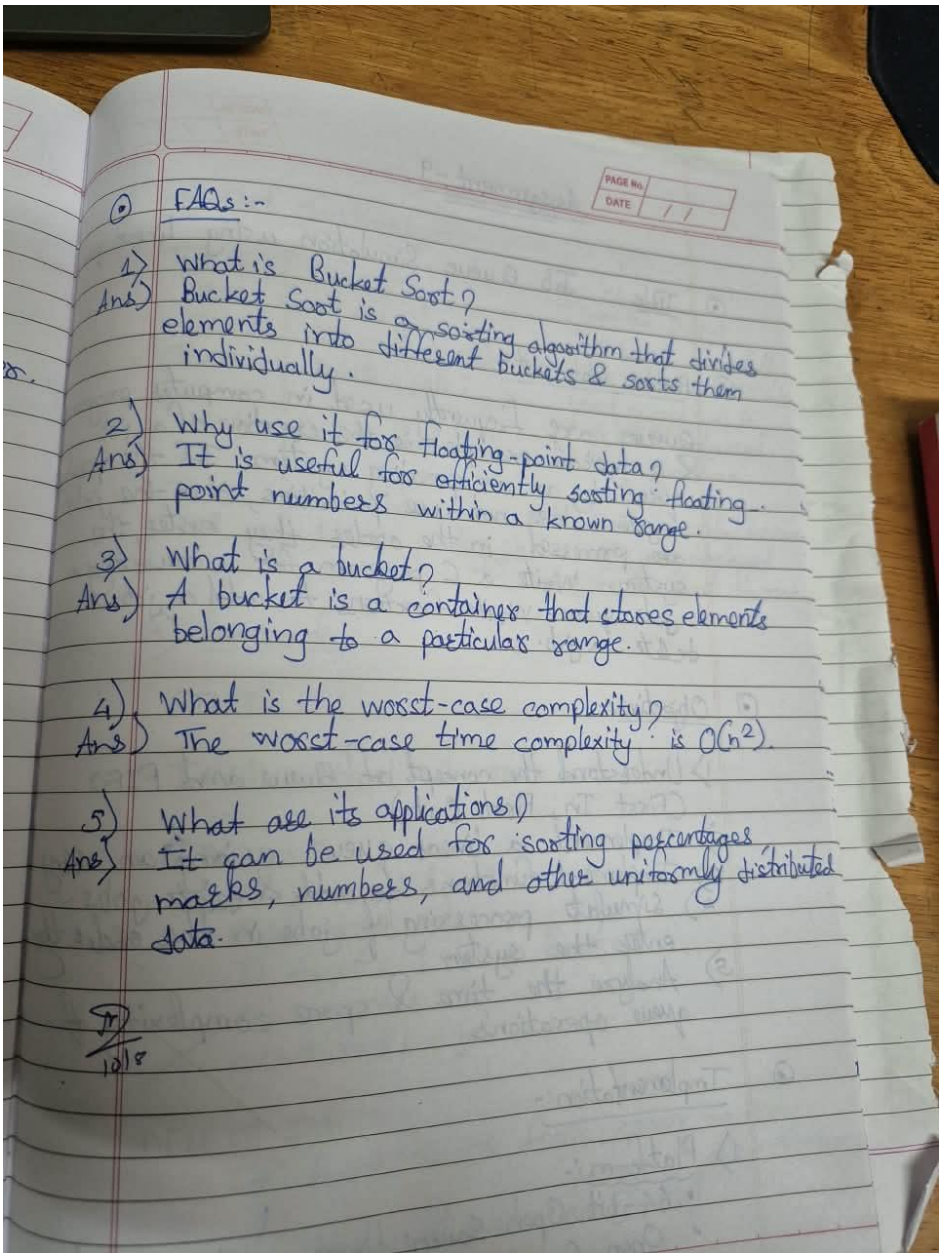
⊙ Time Complexity:-

- Best/Average Case :- $O(n+k)$
- Worst Case : $O(n^2)$

n = numbers of students
k = number of buckets

⊙ Conclusion:-

Bucket Sort efficiently sorts uniformly distributed floating-point percentages & helps identify the top five performers.



⊙ FAQs :-

1) What is Bucket Sort?
Ans) Bucket Sort is a sorting algorithm that divides elements into different buckets & sorts them individually.

2) Why use it for floating-point data?
Ans) It is useful for efficiently sorting floating-point numbers within a known range.

3) What is a bucket?
Ans) A bucket is a container that stores elements belonging to a particular range.

4) What is the worst-case complexity?
Ans) The worst-case time complexity is $O(n^2)$.

5) What are its applications?
Ans) It can be used for sorting percentages, marks, numbers, and other uniformly distributed data.

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